

“EVERYONE With A SMARTPHONE Now Has A Personal SATELLITE NAVIGATION SYSTEM”

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From providing logistic solutions to guiding users, navigation satellite systems are fundamental to modern technologies like the Internet of Things (IoT). Jim Lin, marketing director, SkyTraQ, discusses the operation and scope of GPS/GNSS technology for industrial and consumer navigation and tracking applications in an interaction with Ayushee Sharma

Q. What are the applications of GNSS receivers?

A. Popular Global Navigation Satellite System (GNSS) receiver applications are navigation with smartphones, calling taxi knowing where it is and when it'll arrive, fleet management, vehicle anti-theft GPS locator, GPS sports watch, etc. Less well known are time synchronisation and frequency reference application, which can be used for synchronising time across the globe to the accuracy of 10nsec, or used in GPS Disciplined Oscillator (GPSDO) to generate precise reference frequency for base stations.

Q. How are compatibility and interoperability among different service providers of navigation satellite systems maintained?

A. The compatibility and interoperability of different global and regional navigation systems were carried out bilaterally in the past between US-India on GPS/IRNSS, EU-US on Galileo/GPS, EU-China on Galileo/BeiDou, and US-Japan on GPS/QZSS. Multilaterally, the compatibility and interoperability are discussed through ICG (International Committee on Global Navigation Satellite Systems).

Q. Can you explain the working of GNSS?

A. The principle of GNSS is triangulation. The GNSS receiver measures the travel time of GNSS signals from satellites to the receiver. Using decoded data from GNSS signals, the receiver can compute the position of satellites. By using travel time measurements, one could derive travelled distance by multiplying travel time by the speed of light. By combining positions of three satellites, one can draw three

spheres in 3D, with sphere centre being the satellite position and the sphere radius being travelled distance, the receiver position will be one of the two intersecting points; that is, using three satellite measurements to derive the (x,y,z)- three unknowns of the receiver position.

GNSS receivers typically do not have super-accurate atomic clocks as in the navigation satellites but have low-cost time reference that has some time difference (dt) from the atomic clock. This unknown dt time difference requires another satellite measurement. Once dt is known, the GNSS receiver can adjust its local time to match atomic clock time within 10nsec.

Q. On what factors is the selection of a GNSS module based?

A. Applications such as autonomous ground vehicle precision guidance or UAV to spray pesticides will require centimetre-level-accuracy RTK receivers. Applications that are often in a partial signal blockage environment may benefit from the latest quad-GNSS capable receivers. Being able to use more satellite constellations, a quad-GNSS receiver can have more satellite signals to work within urban canyons than a regular GPS/GLONASS receiver, thus providing better accuracy.

Another consideration may arise from regulation requirements. For example, the AIS-140 application in India requires a minimum of NavIC/GPS capability.

Q. What considerations in design can help ensure GPS accuracy?

A. There are various new developments on the satellite signals to improve GNSS receiver accuracy. Precise Point Positioning (PPP) cor-

rection data is being broadcast on the satellite signal. Japan's QZSS satellites are broadcasting LEX signals to provide centimetre-level accuracy positioning to Japan and Asia-Pacific regions. Galileo plans to provide high accuracy service by broadcasting E6 signals to enable better than 20cm accuracy positioning globally, and so on.

New GNSS chipsets will accommodate these new signals to enable the standalone centimetre-level to decimetre-level accuracy. By using an inertial measurement unit with a GNSS receiver, high accuracy can be maintained for a while after signal blockage to ensure reliability.

Q. Which GNSS system is most often found in smartphones?

A. A smartphone has a minimum of GPS/GLONASS signal receiving capability. Newer models can receive additional Galileo and BeiDou signals. They have similar 2.5m CEP accuracy (inferior to precise RTK or PPP solutions) under the open sky. The ones capable of using more satellite constellations can have better accuracy under adverse signal environments.

Q. How do you see the growth of the consumer satellite navigation market in the coming years?

A. Everyone with a smartphone now has a personal satellite navigation system. New growth areas would be smartwatches with GNSS function for accurate fitness tracking of running distance and pace rate, cars with satellite navigation systems having inertial measurement units, and barometric pressure sensors that provide improved position accuracy in all road environments, giving correct navigation instructions. **EFY**